

Optimization of inflow waveform phase-difference for minimized total cavopulmonary power loss.

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The Fontan operation is a palliative surgical procedure performed on children, born with congenital heart defects that have yielded only a single functioning ventricle. The total cavo-pulmonary connection (TCPC) is a common variant of the Fontan procedure, where the superior vena cava (SVC) and inferior vena cava (IVC) are routed directly into the pulmonary arteries (PA). Due to the limited pumping energy available, optimized hemodynamics, in turn, minimized power loss, inside the TCPC pathway is required for the best optimal surgical outcomes. To complement ongoing efforts to optimize the anatomical geometric design of the surgical Fontan templates, here, we focused on the characterization of power loss changes due to the temporal variations in between SVC and IVC flow waveforms. An experimentally validated pulsatile computational fluid dynamics solver is used to quantify the effect of phase-shift between SVC and IVC inflow waveforms and amplitudes on internal energy dissipation. The unsteady hemodynamics of two standard idealized TCPC geometries are presented, incorporating patient-specific real-time PC-MRI flow waveforms of "functional" Fontan patients. The effects of respiration and pulsatility on the internal energy dissipation of the TCPC pathway are analyzed. Optimization of phase-shift between caval flows is shown to lead to lower energy dissipation up to 30% in these idealized models. For physiological patient-specific caval waveforms, the power loss is reduced significantly (up to 11%) by the optimization of all three major harmonics at the same mean pathway flow (3 L/min). Thus, the hemodynamic efficiency of single ventricle circuits is influenced strongly by the caval flow waveform quality, which is regulated through respiratory dependent physiological pathways. The proposed patient-specific waveform optimization protocol may potentially inspire new therapeutic applications to aid postoperative hemodynamics and improve the well being of the Fontan patients.

Fontan-associated liver disease: the importance of multidisciplinary teamwork in its management.

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The Fontan operation, which directly connects the superior and inferior vena cava to the pulmonary artery, is a palliative surgery for children with a functional or anatomic single ventricle. This procedure leads to hemodynamic changes (Fontan circulation) in patients, who tend to develop congestive hepatic fibrosis characterized by sinusoidal fibrosis and dilatation beginning approximately 10 years after the procedure. In addition, in the context of severe fibrosis and cirrhosis, hepato-gastrointestinal complications including hepatocellular carcinoma, focal nodular hyperplasia, and portal hypertension can arise. Fontan-associated liver disease (FALD) encompasses the broad spectrum of liver alterations secondary to postoperative hemodynamic changes, and the effective management of FALD requires contributions from specialists in hepatology, gastroenterology, surgery, radiology, histopathology, and pediatric and adult cardiology. In this article, we outline the pathogenesis of FALD and discuss the importance of a multidisciplinary collaborative approach to its management.

Laparoscopic hysterectomy for endometrial cancer after the Fontan procedure: The first case report.

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The Fontan procedure is a palliative cardiac surgery performed for children with a single functional ventricle, and its aim is to divert central venous return to the pulmonary artery without passing through the right ventricle. We herein report the first case of endometrial cancer after the Fontan procedure that was successfully treated with laparoscopic hysterectomy. A 38-year-old female who underwent the Fontan procedure at the age of 23 presented with abnormal genital bleeding and was diagnosed with endometrial cancer. Computed tomography revealed distinctive venous vasculature in the pelvis: most of the venous return from the left lower extremity occurred via the extremely distended left ovarian vein. Laparoscopic total hysterectomy and bilateral salpingo-oophorectomy were performed. The pneumoperitoneum was maintained at 8-10 mmHg during the operation so that the high intra-abdominal pressure would not interfere with venous return. In addition, the left ovarian vein was first test clamped to ensure that circulation was maintained and was then resected. Although the volume of blood loss reached 358 ml due to high venous pressure, her postoperative course was favorable, and she had no signs of recurrence at the 12-month follow-up. Our case suggests that laparoscopic hysterectomy might be safe and feasible for patients with endometrial cancer after the Fontan procedure, as long as a preoperative study of pelvic vascularization and intraoperative monitoring of the circulation is carried out.

Cross-sectional imaging of the Fontan circuit in adult congenital heart disease.

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The Fontan circuit is the result of a palliative surgical procedure that is performed in patients with a functionally single ventricle cardiac anomaly. The success of this operation has resulted in an increasing population of adults with this anatomy and physiology. The purpose of this article is to familiarize the general radiologist with the expected anatomy and cross-sectional imaging findings, highlight special imaging considerations, and examine the common complications that are encountered in this group of patients.

Initial syncope associated with alternating attacks of supraventricular tachycardia and atrioventricular block long after surgical correction of tricuspid atresia.

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The Fontan procedure has become a generic term to define a surgical procedure that orients the systemic venous return directly to the pulmonary arteries, and has been used as a palliative operation for pulmonary atresia, tricuspid atresia, double inlet ventricle and complex single ventricle. The earliest type of Fontan procedure was a simple atriopulmonary anastomosis between the right atrium and the pulmonary artery. Atrial arrhythmias, particularly atrial flutter and sinus node dysfunction may occur in the early and late postoperative period after simple atriopulmonary anastomosis. The case presented here represents a much delayed occurrence of an initial syncope due to alternating attacks of SVT (supraventricular tachycardia) and second degree heart block on admission 21 years after simple atriopulmonary anastomosis performed for the correction of tricuspid atresia.

Caesarean section following the Fontan procedure: two different deliveries and different anaesthetic choices in the same patient.

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The Fontan procedure is performed for patients with a hypoplastic right ventricle, and pregnancies following this palliative surgery are likely to increase. We present a parturient with the Fontan physiology who successfully underwent two consecutive caesarean deliveries; the first under general anaesthesia for emergency surgery and the second under regional anaesthesia for elective surgery. We suggest that pregnancy and delivery do not typically adversely affect maternal cardiac status in these patients. Attention must be paid, however, to fetal loss, prematurity, growth retardation and associated cardiac congenital malformations for which insufficient data exist in the literature in this patient population.

[Followup of adults with a Fontan-type cavopulmonary derivation].

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The Fontan procedure is a technique used in single ventricle syndromes and initiates a circulation which bypasses the right ventricle. It was described by Fontan in 1971. Many variations of this operation have been described and classified under the term of total cavopulmonary derivation. These operations result on long-term functional benefit. However, the 15 year survival rate is only 55 to 80%. This procedure (often carried out in childhood) is considered to be palliative rather than curative. The mortality is due to long-term complications, the frequency of which increases with the length of follow-up of the patient. The causes are: the Fontan circulation itself (arrhythmias, thrombosis, inferior caval stasis (hepatic, exsudative enteropathy), recurrence of cyanosis); the underlying cardiac disease (ventricular failure, atrioventricular valvular regurgitation); other complications (endocarditis and extracardiac, in particular orthopedic problems). Any of these complications may be an indication for cardiac transplantation, a high risk procedure because of previous multiple surgical interventions.

Role of imaging in the evaluation of single ventricle with the Fontan palliation.

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The Fontan operation for single ventricle palliation consists of the creation of a complete cavopulmonary connection, usually by incorporating inferior vena caval flow into a pulmonary arterial circulation already receiving flow from the superior vena cava. In single ventricle palliated in this way, the anatomy is complex, and the pathophysiological complications are frequent; so, cardiac imaging is a key aspect of clinical surveillance. Common problems that echocardiography and MRI may disclose and characterise in the Fontan palliation of single ventricle include obstruction of systemic venous and pulmonary arterial flow, atrioventricular and semilunar valve dysfunction, unintended collateral flow patterns, ventricular dysfunction, aortic arch obstruction, interatrial obstruction, fenestration flow and patch leaks. Despite the broad scope of these modalities for detection of such problems, often no single imaging method is comprehensive in any given patient. Therefore, physicians must recognise the limitations of each modality, and circumvent these by application of suitable alternatives. New imaging tools are becoming available, which may ultimately prove to be of value in the Fontan circulation. Proper application of diverse new technologies such as four dimensional flow, computational fluid dynamics and

three-dimensional printing will require critical evaluation in the single ventricle population.

Pearls and Pitfalls in Pediatric Fontan Operation Imaging.

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The Fontan operation or the total cavopulmonary connection is a palliative surgery for single ventricle congenital heart disease where the systemic venous return circumvents a pumping chamber and flows directly into the pulmonary circuit. With surgical and medical advances, there has been improvement in life expectancy of these patients, however, it has also resulted in unique complications from the physiology that requires diligent surveillance. A critical component relies on optimal imaging for diagnosis and treatment of these complications. This article describes the normal anatomy of the Fontan circulation, current imaging modalities and techniques, and frequently encountered complications seen when imaging the patients who have undergone Fontan palliation.

Predicting Hemodynamic Performance of Fontan Operation for Glenn Physiology using Computational Fluid Dynamics: Ten Patient-specific Cases.

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Single ventricle hearts have only one ventricle that can pump blood effectively and the treatment requires three stages of operations to reconfigure the heart and circulatory system. At the second stage, Glenn procedure is performed to connect superior vena cava (SVC) to the pulmonary arteries (PA). For the third and most complex operation, called Fontan, an extracardiac conduit is used to connect inferior vena cava (IVC) to the PL and thereafter no deoxygenated blood goes to the heart. Predicting Hemodynamic Performance of Fontan Operation using computational fluid dynamics (CFD) is hypothesized to improve outcomes and optimize this treatment planning in children with single-ventricle heart disease. An important reason for this surgical planning is to reduce the development of pulmonary arteriovenous malformations (PAVM) and the need to perform Fontan revisions. The purpose of this study was to develop a model for Fontan surgical planning and use this model to compare blood circulation in two designed graft types of Fontan operation known as T-shape and Y-graft. The functionality of grafts was compared in terms of power loss (PL) and hepatic flow distribution (HFD), a known factor in PAVM development. To perform this study, ten single-ventricle children with Glenn physiology were included and a CFD model was developed to estimate the blood flow circulation to the left and right pulmonary arteries. The estimated blood flow by CFD was compared with that measured by cardiovascular magnetic resonance. Results showed that there was an excellent agreement between the net blood flow in the right and left pulmonary arteries computed by CFD and CMR (ICC= 0.98, P-value ≥ 0.21). After validating the accuracy of each CFD model, Fontan operations using T-shape and Y-graft conduits were performed in silico for each patient and the developed CFD model was used to predict the post-surgical PL and HFD. We found that the PL in the Y-graft was significantly lower than in the T-shape (P-value ≤ 0.001) and HFD was significantly better balanced in Y-graft compared to the T-shape (P-value=0.004).